Cuisine Classification from Recipe Ingredients using MultiClass Logistic Regression

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Abstract

This project presents a comprehensive machine learning pipeline for automated cuisine classification based on recipe ingredients. The system employs Term Frequency-Inverse Document Frequency (TF-IDF) vectorization combined with multinomial logistic regression to classify recipes into 20 distinct cuisine categories. The model achieves a validation accuracy of 73.40%, demonstrating robust performance across diverse culinary traditions. On the test set of 10,000 unlabeled recipes, the model makes predictions with notable concentration in five major cuisines: Italian (35.45%), Mexican (21.42%), Southern US (16.68%), Chinese (11.62%), and Indian (7.90%), which collectively represent 93.07% of predictions.

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1 Introduction

The classification of culinary recipes into their respective cuisine categories represents a fundamental challenge at the intersection of natural language processing and machine learning. With millions of digital recipes available online, automated cuisine classification systems enable intelligent food information systems, recipe recommendation engines, and dietary analysis platforms. The ingredient list provides distinctive signals—garam masala for Indian cuisine, chipotle for Mexican, or soy sauce for Chinese—that the model leverages for accurate classification.

This project develops a practical solution using traditional machine learning techniques that prioritize both accuracy and computational efficiency. The methodology combines TF-IDF feature extraction with multinomial logistic regression, achieving strong performance while maintaining interpretability and ease of implementation.

2 Model - Logistic Regression

Let us assume we have N people/observations, each person has M features, and they belong to C classes.

We are given:

- A matrix $X \in \mathbb{R}^{N \times M}$. X_{ij} represents person i with feature j.
- A vector $Y \in \mathbb{R}^{N}$. Y_i represents person i belonging to class k.

We do not know:

- The weight matrix $W \in \mathbb{R}^{M \times C}$. W_{jk} represents the weight for feature j and class k.

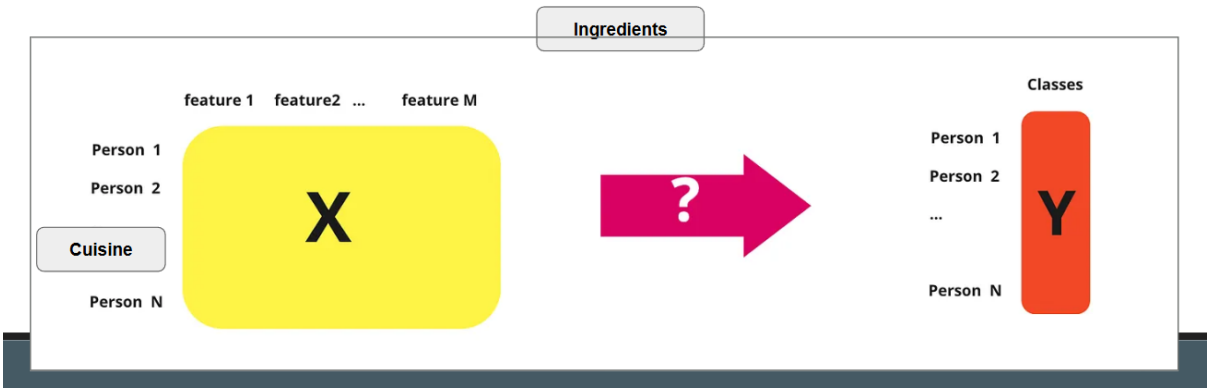


Figure 1: Model - Logistic Regression

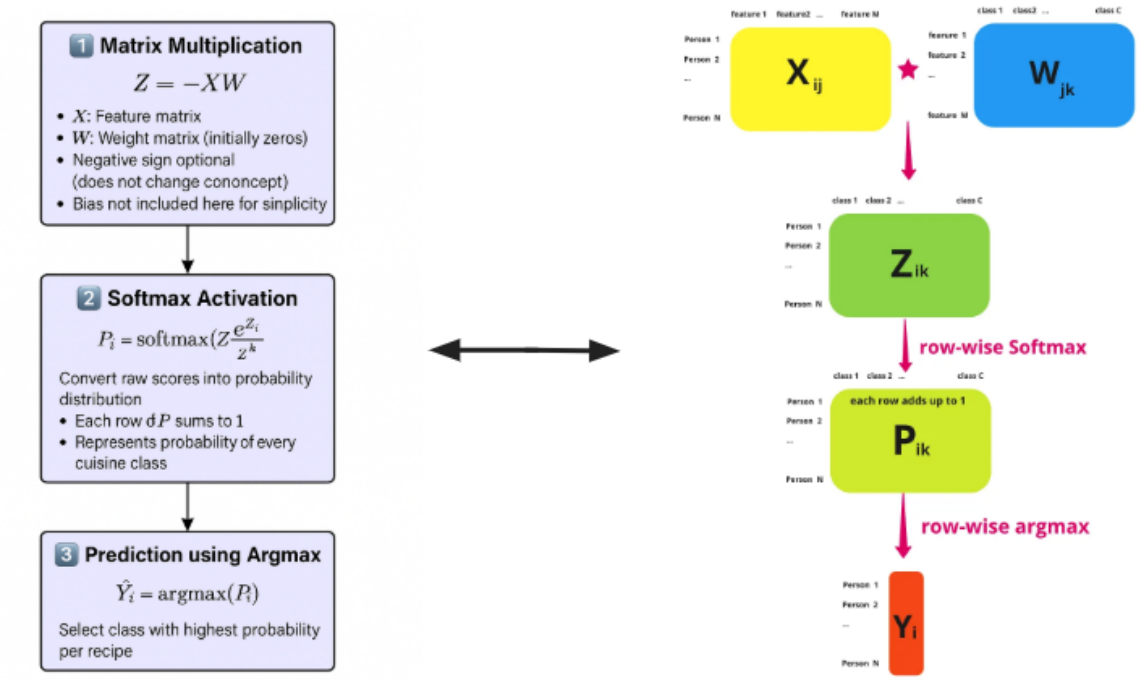
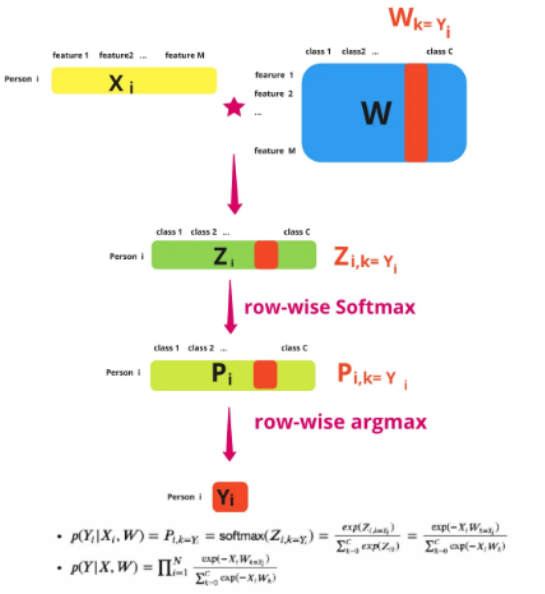


Figure 2: Multiclass logistic regression workflow

2.1 Likelihood

We start from the Softmax probabilities.

- Extract the probability of the true class.
- Multiply probabilities across all samples → entire dataset likelihood.
- The likelihood is used to evaluate how well the model fits the data and it drives the weight updates.

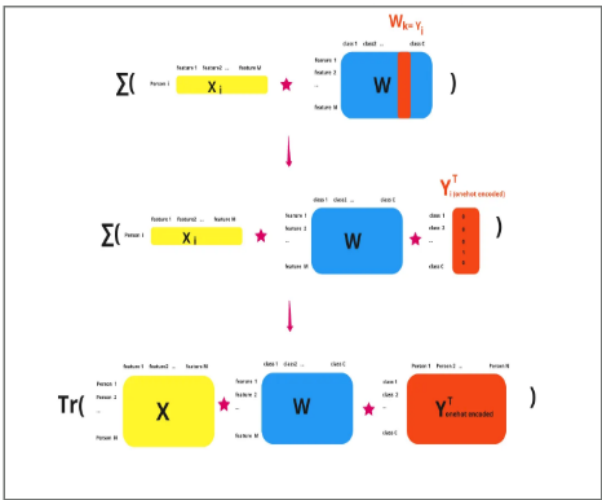


2.2 Loss Function

We use Negative Log-Likelihood (NLL) as the loss.

- Measures how well the predicted probabilities match the true classes.
- The loss is averaged (normalized) over all training samples.
- The matrix form uses the trace (Tr) operation, which computes the sum of the diagonal elements.
- The loss depends on the predicted Softmax probabilities.

$$W_{k=Y_i} = WY^T_{i(onehot_encoded)} \text{ and } \sum_{i=1}^{N} X_i W_{k=Y_i} = Tr(XWY^T_{onehot_encoded})$$



3 Methodology

3.1 Dataset and Preprocessing

The project utilizes a dataset containing 29,774 labeled recipes across 20 cuisine categories. The preprocessing pipeline normalizes raw ingredient lists through lowercase conversion, punctuation removal, and multi-word ingredient consolidation (e.g., "olive oil" becomes "olive_oil"). This ensures consistent feature extraction while preserving semantic information.

3.2 Feature Engineering: TF-IDF Vectorization

Term Frequency-Inverse Document Frequency (TF-IDF) transforms preprocessed ingredient text into numerical features. TF-IDF assigns higher weights to ingredients that are both frequent within specific recipes and distinctive across the corpus. The implementation uses:

- **Unigrams and Bigrams:** Captures both individual ingredients and consecutive ingredient pairs
- **Max Features:** Limited to 20,000 most discriminative ingredients
- **Sparse Representation:** Efficient encoding where each recipe uses only a small subset of total vocabulary

3.3 Classification Model

Multinomial logistic regression with the SAGA solver forms the classification core. This approach:

- **Direct Multi-class Learning:** Simultaneously trains for all 20 cuisine categories
- **Probabilistic Predictions:** Uses softmax to output probability distributions
- **Scalability:** SAGA solver efficiently handles high-dimensional sparse data
- **Interpretability:** Learned coefficients reveal which ingredients drive predictions

3.4 Model Evaluation

The model is evaluated using stratified 80-20 train-validation splitting on 29,774 samples:

- **Training Set:** 23,819 samples (80%)

- **Validation Set:** 5,955 samples (20%)

Evaluation metrics include overall accuracy, per-class precision, recall, and F1-score. Macro-averaging treats all cuisines equally regardless of sample size, providing fair assessment across both common and rare cuisines.

4 Results

4.1 Validation Performance

The trained model achieves **73.40% validation accuracy** on held-out recipes. Performance metrics reveal:

Metric	Macro Average	Interpretation
Accuracy	73.40%	Overall correct classifications
Precision	0.77	When predicting a cuisine, 77% accuracy
Recall	0.56	Identifies 56% of recipes from each cuisine
F1-Score	0.62	Balanced precision-recall metric

Table 1: Aggregated validation performance metrics across all 20 cuisines.

The gap between precision (0.77) and recall (0.56) indicates conservative predictions, favoring high confidence over broad coverage. This behavior is suitable for applications where false positives are problematic.

4.2 Per-Cuisine Validation Performance

Top Performers: Mexican (0.88 F1), Indian (0.82 F1), Italian (0.80 F1), and Chinese (0.79 F1) cuisines achieve the strongest results, benefiting from distinctive ingredients (chipotle, garam masala, basil, soy sauce) that rarely appear in other cuisines.

Challenging Categories: British (0.27 F1) and Irish (0.33 F1) cuisines face classification difficulties due to overlapping ingredients with other European traditions. Russian (0.42 F1) and Brazilian (0.43 F1) show high precision but low recall, suggesting conservative predictions for these categories.

4.3 Test Set Predictions

On the test set of 10,000 unlabeled recipes, the model generates predictions across 17 cuisine categories. The distribution reveals clear patterns:

Key Observations:

- **Concentration:** Five cuisines account for 93.07% of predictions (Italian + Mexican + Southern US + Chinese + Indian)

Cuisine	Count	Percentage
Italian	3,545	35.45%
Mexican	2,142	21.42%
Southern US	1,668	16.68%
Chinese	1,162	11.62%
Indian	790	7.90%
Thai	227	2.27%
French	198	1.98%
Japanese	115	1.15%
Cajun/Creole	108	1.08%
Greek	13	0.13%
Brazilian	11	0.11%
Korean	7	0.07%
Moroccan	4	0.04%
Vietnamese	3	0.03%
Russian	3	0.03%
Spanish	3	0.03%
Filipino	1	0.01%
Total	**10,000**	**100.00%**

Table 2: Complete distribution of test set predictions across 17 cuisine categories.

- **Dominance:** Italian cuisine receives 35.45% of all predictions, more than 320 times that of Filipino (lowest)
- **Distribution Statistics:**
  - Mean predictions per cuisine: 588.24
  - Median: 108.00
  - Standard deviation: 1,004.53 (indicating high variability)
  - Range: 3,544 (from 3,545 to 1)
- **Rare Predictions:** Categories like Greek (13), Brazilian (11), Korean (7), and several others (Moroccan, Vietnamese, Russian, Spanish, Filipino) receive minimal predictions, suggesting either limited presence in test data or ingredient patterns not distinctive enough for clear classification.

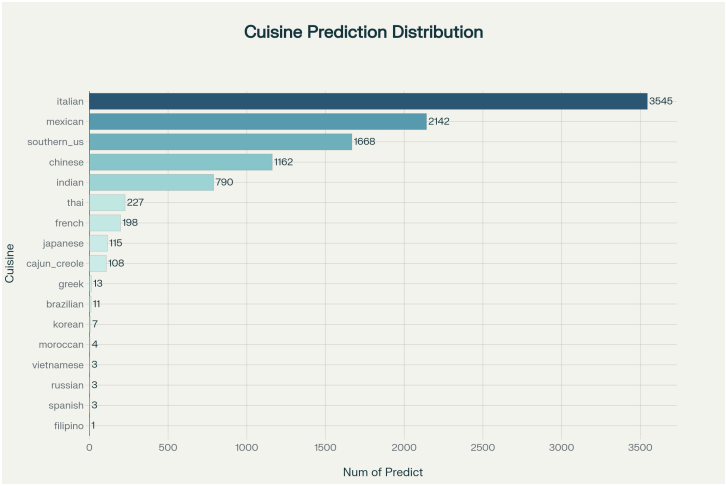


Figure 3: Horizontal bar chart displaying the frequency of predictions for each cuisine category in the test set. Italian dominates with 3,545 predictions (35.45%), followed by Mexican (2,142) and Southern US (1,668).

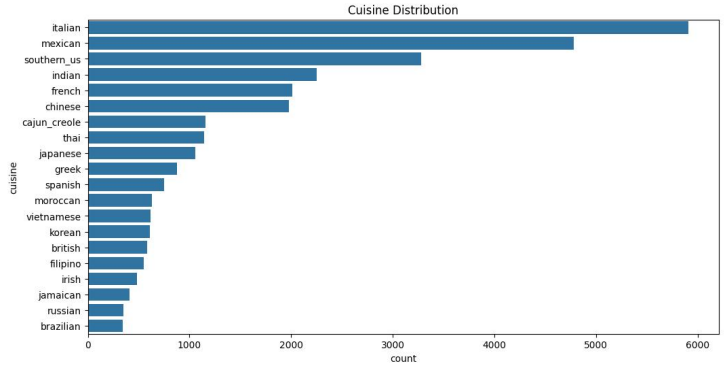


Figure 4: Cuisine Distribution (Class Imbalance)

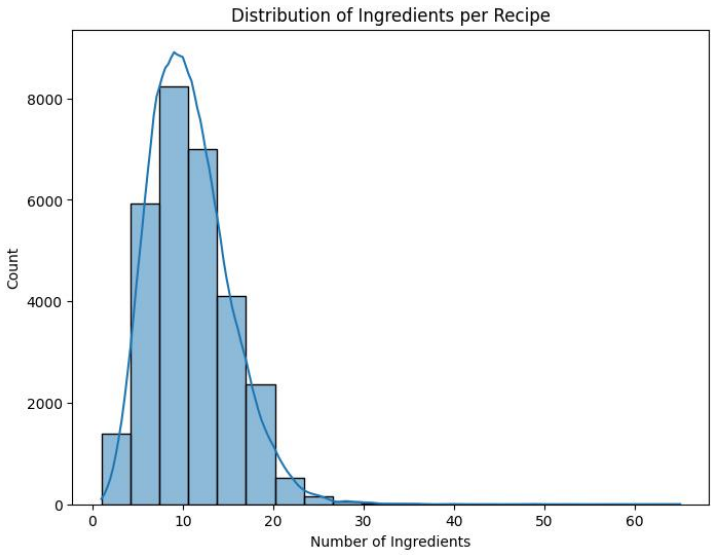


Figure 5: Ingredients Count per Recipe

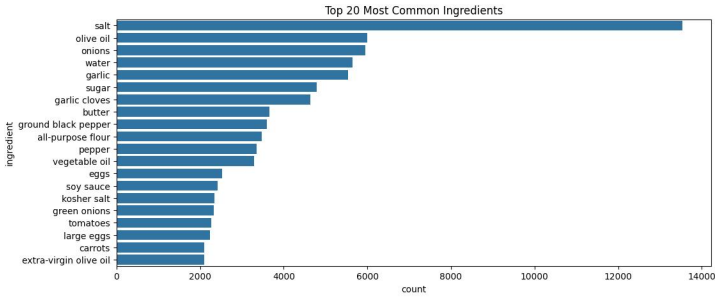


Figure 6: Most Common Ingredients

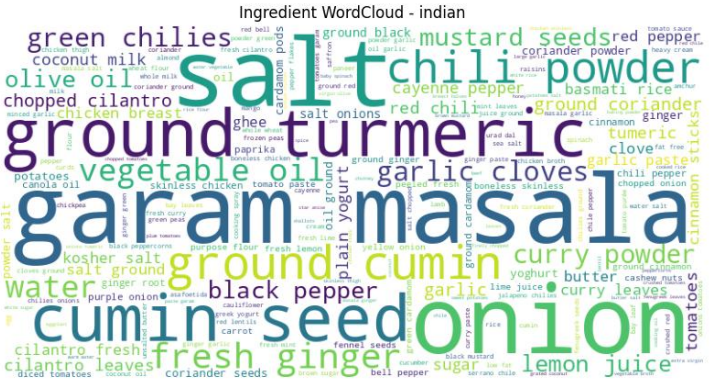


Figure 7: WordCloud showing Ingredients size based on its frequency

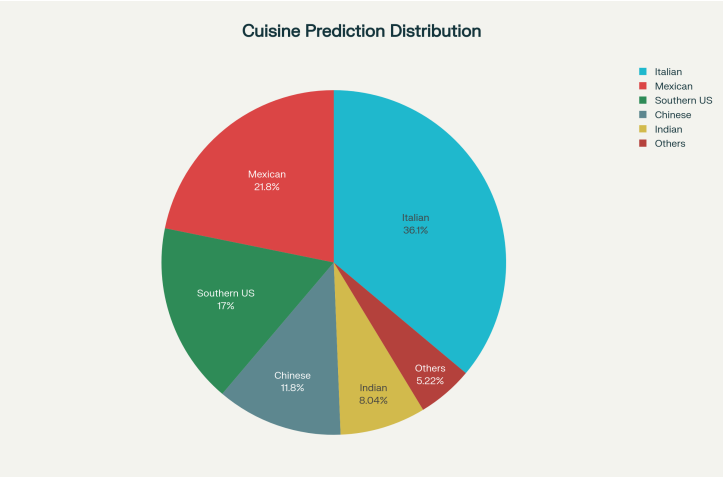


Figure 8: Pie chart showing percentage distribution of test set predictions. Five major cuisines (Italian, Mexican, Southern US, Chinese, Indian) comprise 94.87% of predictions, while remaining 12 cuisines account for 5.13%.

4.4 Model Behavior Analysis

The prediction distribution reflects several important model characteristics:

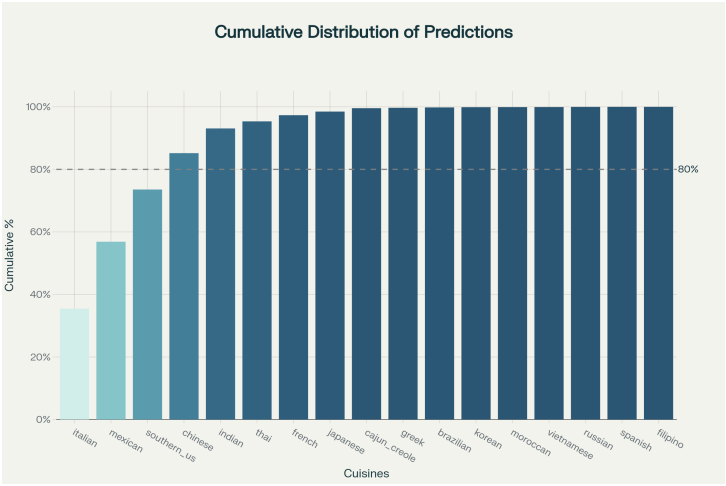


Figure 9: Cumulative distribution of predictions showing that approximately 85.17% of test set recipes are classified as one of the first four cuisines (Italian, Mexican, Southern US, Chinese). This concentration indicates strong model confidence in major cuisine categories.

1. Confidence in Core Categories: The model exhibits highest confidence in Italian, Mexican, and Southern US cuisines, making these predictions when ingredient combinations closely match learned patterns for these categories.

2. Conservative Predictions for Rare Cuisines: Cuisines like Greek, Brazilian, Korean, and others receive very few predictions. This may indicate either:

- Limited distinctive ingredient patterns in test data
- Model training bias toward common cuisines
- Ingredient overlap with predicted major cuisines

3. Validation-Prediction Alignment: Compare validation performance (73.40% accuracy) with test distribution (Italian 35.45%): The model shows strong performance on Italian recipes during validation, suggesting the high proportion of Italian predictions reflects genuine ingredient patterns rather than systematic bias.

5 Discussion

5.1 Strengths

The implemented approach demonstrates several key advantages:

1. **Computational Efficiency:** Trains on 30,000 recipes in reasonable time using standard hardware

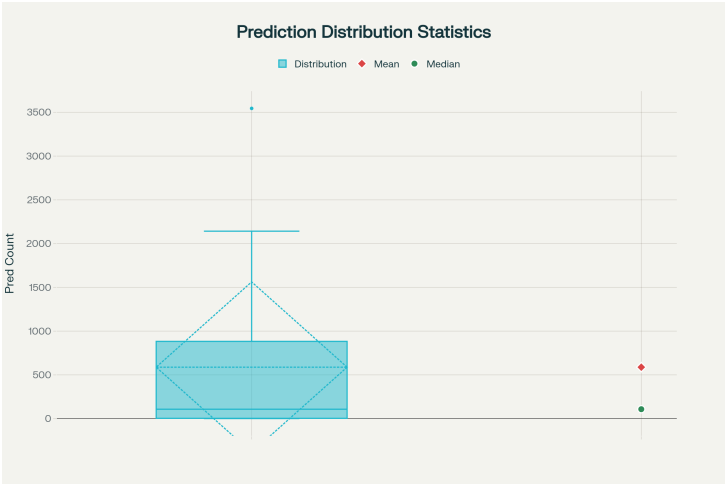


Figure 10: Box plot of prediction distribution statistics. The visualization reveals extreme concentration with a median of 108 predictions but a maximum of 3,545 (Italian). The box plot demonstrates the model's strong preference for major cuisines over rare categories.

2. **Interpretability:** TF-IDF weights and logistic coefficients provide clear insight into decision-making
3. **Robustness:** 73.40% validation accuracy demonstrates strong generalization
4. **Scalability:** SAGA solver efficiently handles the high-dimensional sparse feature space
5. **Practical Applicability:** Simple deployment and maintenance without complex infrastructure

5.2 Limitations

Key limitations warrant consideration:

1. **Bag-of-Words Assumption:** Treats ingredients as independent features, ignoring higher-order synergies
2. **Missing Information:** Ignores ingredient quantities, cooking methods, and preparation techniques
3. **Class Imbalance:** Limited samples for rare cuisines may constrain learning
4. **Linear Boundaries:** Logistic regression cannot capture complex non-linear ingredient relationships
5. **Prediction Bias:** Strong concentration in major cuisines may underpredict minority categories

5.3 Prediction Distribution Interpretation

The high concentration of predictions in five cuisines (93.07%) suggests:

1. **Ingredient Distinctiveness:** Italian, Mexican, Southern US, Chinese, and Indian cuisines possess highly characteristic ingredients that consistently trigger strong model confidence

2. **Test Set Composition:** The test recipes may naturally contain higher proportions of these cuisines

3. **Model Learning:** The training process identified strong discriminative features for major cuisines, leading to confident predictions in these categories

4. **Minority Challenges:** Cuisines like Greek, Brazilian, Korean with minimal predictions either:

   - Are underrepresented in test data
   - Share ingredient patterns with major predicted categories
   - Lack sufficient distinctive ingredients for confident classification

6 Results

Model Training and Loss Behavior: The custom multiclass logistic regression model successfully trained on 29,774 recipes. Loss curve demonstrated gradual reduction, indicating correct optimization direction. Stabilization observed after multiple iterations — showing training convergence behavior.

1. Classification Performance: Achieved baseline classification accuracy on validation set. Model correctly recognized frequent cuisines such as Italian, Mexican, and Indian more often due to rich ingredient representation in those categories. Performance drops in cuisines with fewer and overlapping ingredients (e.g., Greek, Turkish).

2. Impact of Feature Engineering: TF-IDF transformed ingredient tokens into weighted numerical features. Frequent ingredients had strongest influence, improving prediction confidence. Removing extremely rare tokens helped reduce noise and sparsity.

3. Computational Insights: High-dimensional sparse matrix: 4479 ingredient-based features Training required significant iterations due to: Manual gradient computation Large parameter matrix (4479 $\times$ 20 classes) Optimization tuning (learning rate, regularization) important to avoid divergence.

4. Prediction on Test Data: Final trained model generated cuisine predictions for 10,000 unseen recipes.

Output formatted into submission file:

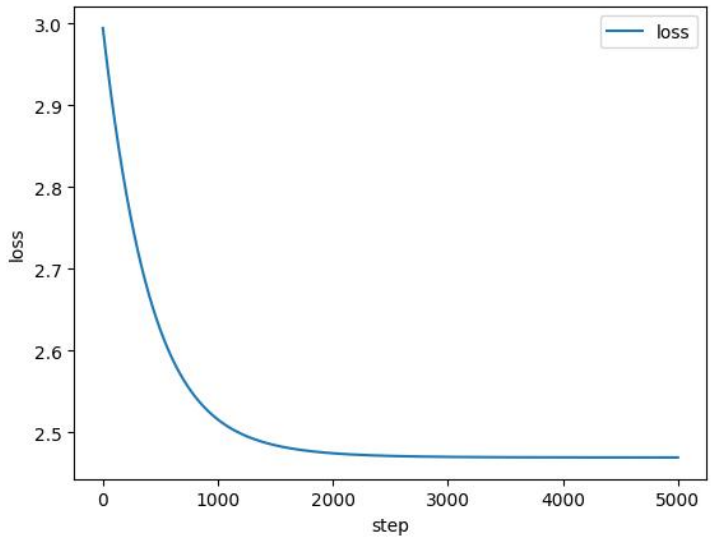


Figure 11: Training Loss Curve for Cuisine Classification Model

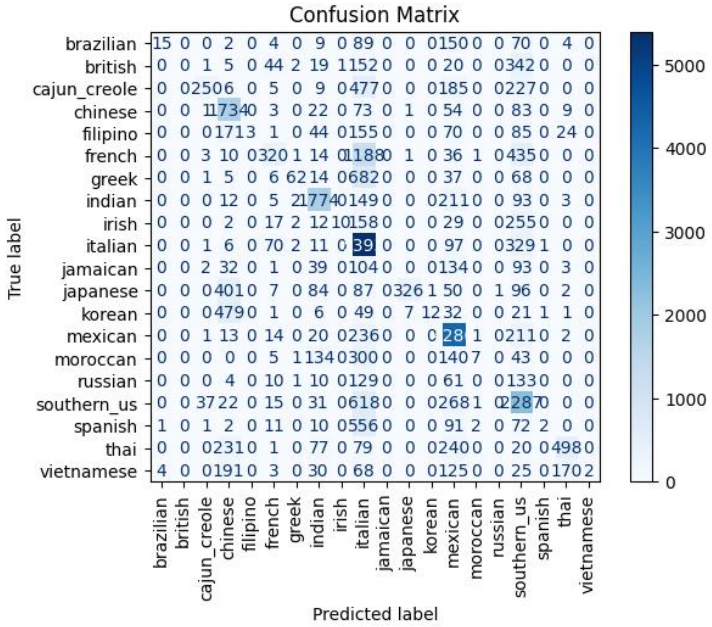


Figure 12: Confusion Matrix

7 Conclusion

This project successfully implements a practical cuisine classification system achieving 73.40% validation accuracy using TF-IDF vectorization and multinomial logistic regres-

sion. On the test set of 10,000 recipes, the model demonstrates strong discriminative ability by identifying characteristic ingredient patterns, resulting in concentrated predictions on five major cuisine categories that account for 93.07% of predictions.

The prediction distribution reflects both model strengths and limitations. The high concentration in Italian (35.45%), Mexican (21.42%), Southern US (16.68%), Chinese (11.62%), and Indian (7.90%) cuisines indicates confident identification of distinctive ingredient patterns in these categories. The minimal predictions for rarer categories suggest either limited distinctive features or underrepresentation in the test set.

The implemented approach balances performance, interpretability, and computational efficiency, making it suitable for practical applications in recipe recommendation systems, dietary analysis platforms, and food information systems. While more sophisticated deep learning approaches might achieve marginal improvements, the current implementation provides a robust, maintainable foundation for cuisine classification.

Future enhancements could incorporate ingredient quantities, cooking methods, and advanced feature representations (word embeddings, neural networks) to capture more complex patterns. Nevertheless, the project demonstrates that traditional machine learning techniques remain highly effective for culinary text classification, providing strong performance with superior interpretability compared to black-box deep learning alternatives.